Classification Shifting in the Cash Flow Statement: Evidence from India

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Classification shifting in the cash flow statement

Particulars	*Amount*
Cash Flow from Operating Activities	
Inflows	
Outflows	
Cash Flow from Investing Activities	
Inflows	
Outflows	
Cash Flow from Financing Activities	
Inflows	
Outflows	

Research questions

- Do managers of Indian firms misclassify cash flows in order to inflate operating cash flows?
  - *Yes, techniques*
- Are financially distressed firms in India more likely to misclassify cash flows in order to inflate operating cash flows?
  - *Yes*

Research questions

- Is the magnitude of such misclassification higher for the firms in India as compared to those in the United States?
  - *Yes*
- Is the magnitude of such misclassification higher for the financially distressed firms in India as compared to those in the United States?
  - *Yes*

Techniques

- Investing cash inflows to operating cash inflows
  - Proceeds from sale of ‘available-for-sale’ securities (Nautica Enterprises Inc.)
- Financing cash inflows to operating cash inflows
  - Change in notes payable for vehicle inventory purchased from a manufacturer *unaffiliated* with the lender (Asbury Automotive Group Inc.)

Techniques

- Operating cash outflows to investing cash outflows
  - Expenses paid for sponsorship and newspaper advertisements as a part of Property, Plant and Equipment (HealthSouth Inc.)
- Operating cash outflows to financing cash outflows

Motivation and Hypotheses

Significance of operating cash flows

- More number of firms and analysts now issue cash flow forecasts (DeFond and Hung 2003, DeFond and Hung 2007, Wasley and Wu 2006, Call 2008)
- Investors have started paying more attention to cash flow from operations figure

  (Schilit and Perler 2010)
- Positive stock price reaction when cash flow surprise is positive (DeFond and Hung 2003, Zhang 2007, Brown et al. 2010)

Significance of operating cash flows

- Operating cash flows are considered sustainable and have valuation related consequences (Mulford and Comiskey 2005)
- Debt covenants and cash flow based compensation also motivate managers to indulge in creative cash flow reporting (Mulford and Comiskey 2005, Frankel et al. 2010)

Existing evidence on classification shifting

- Lee (2012)
  - Financial distress
  - Long-term credit rating near the investment/non-investment grade cut-off
  - Analysts' cash flow forecasts
  - Higher association between a firm's stock returns and its cash flow from operations
- Hollie et al. (2011)
  - Restated cash flow from operations is 50% less than the originally reported figure
  - More prevalent in retail and financial sectors

Existing evidence on classification shifting

- Classification shifting is used in the East Asian countries (Haw et al. 2011)
  - Its use intensifies with the presence of controlling shareholders
- Earnings management is more likely in countries with weak legal enforcement and investor protection (Leuz et al. 2003, Haw et al. 2004)
- Lesser regulators' and auditors' scrutiny

  (McVay 2006)

Opportunities

- Weak corporate governance and investor protection (Narayanaswamy et al. 2012)
  - Class-action lawsuits and lawsuits against the auditors are uncommon
  - Monetary penalties are miniscule and enforcement of the laws is also weak
- Serious Fraud Investigation Office (SFIO) and the case of Satyam Computer Services Limited
- Dominant role of family firms and presence of controlling shareholders

Incentives

- Analysts' cash flow forecasts
- About 90% of the Indian firms with analysts' earnings forecasts also have analysts' cash flow forecasts (DeFond and Hung 2007)

Hypothesis 1

H1A: Managers of Indian firms shift investing or financing cash inflows to operating cash inflows.

H1B: Managers of Indian firms shift operating cash outflows to investing or financing cash outflows.

Hypothesis 2

H2A: Managers of Indian firms shift investing or financing cash inflows to operating cash inflows more than the managers of firms in the United States.

H2B: Managers of Indian firms shift operating cash outflows to investing or financing cash outflows more than the managers of firms in the United States.

Impact of financial distress

- Managers of distressed firms consider cash flow as an important performance indicator for outsiders (Graham et al. 2005)
- Cash flows help in predicting bankruptcy (Sharma 2001, Ohlson 1980)
- Firms upward manage reported cash flow from operations when in financial distress

  (Lee 2012)

Hypothesis 3

- Cash flow restatement firms have higher debt ratio (Hollie et al. 2011)

H3A: Managers of financially distressed firms (in India) are more likely to shift investing or financing cash inflows to operating cash inflows.

H3B: Managers of financially distressed firms (in India) are more likely to shift operating cash outflows to investing or financing cash outflows.

Hypothesis 4

- Bankruptcy costs are comparatively lower in India due to weak firm winding up procedures (Narayanaswamy et al. 2012)
- Weak governance

H4: There is an insignificant difference in the magnitude of cash flow misclassification between the financially distressed firms in India and the United States.

Sample selection and descriptive statistics

- BSE listed non-financial Indian firms from 1995-2011
- Data source: Prowess
- 13,305 firm years/1,370 firms
- Data winsorization at extreme 1%
- Sample comprises of bigger firms as compared to population

Sample selection and descriptive statistics

- US Data
  - Compustat North America (Fundamentals Annual)
  - 76,258 firm-years, 1989-2010
- Median operating cash flows (*cfo*): 7.3% of lagged total assets
- Firms in the United States are less financially distressed on average

Research design

- *Stage I*: Estimation of unexpected cash flows (Dechow et al. 1998, Roychowdhury 2006)

$$CFO_{i,t}/A_{i,t-1} = \beta_0 + \beta_1 (1/A_{i,t-1}) + \beta_2 (S_{i,t}/A_{i,t-1}) + \beta_3 (\Delta S_{i,t}/A_{i,t-1}) + \varepsilon_{i,t}$$

Research design

- *Stage II*

$$UE\_CFO_{i,t} = \alpha_0 + \alpha_1\, IVO_{i,t} + \alpha_2\, IVI_{i,t} + \alpha_3\, FINO_{i,t} + \alpha_4\, FINI_{i,t} + \alpha_5\, ZSCORE_{i,t-1}$$

$$+ \alpha_6\, ZSCORE_{i,t-1}{*}IVO_{i,t} + \alpha_7\, ZSCORE_{i,t-1}{*}IVI_{i,t} + \alpha_8\, ZSCORE_{i,t-1}{*}FINO_{i,t}$$

$$+ \alpha_9\, ZSCORE_{i,t-1}{*}FINI_{i,t} + \alpha_{10}\, ROA_{i,t} + \alpha_{11}\, SIZE_{i,t} + \alpha_{12}\, MTB_{i,t} + \alpha_{13}\, DACC_{i,t} + \omega_{i,t}$$

Research design

$$UE\_CFO_{i,t} = \alpha_0 + \alpha_1\,CFF_{i,t} + \alpha_2\,CFI_{i,t} + \alpha_3\,ZSCORE_{i,t-1} + \alpha_4\,ZSCORE_{i,t-1}{*}CFF_{i,t} + \alpha_5\,ZSCORE_{i,t-1}{*}CFI_{i,t} + \alpha_6\,ROA_{i,t} + \alpha_7\,SIZE_{i,t} + \alpha_8\,MTB_{i,t} + \alpha_9\,DACC_{i,t} + \delta_{i,t}$$

Altman's (2002) Z-Score =

6.56*(Working capital$_{t-1}$/Total assets$_{t-1}$) +

3.26*(Retained earnings$_{t-1}$/Total assets$_{t-1}$) +

6.72*(Profit before interest and tax$_{t-1}$/Total assets$_{t-1}$) +

1.05*(Common equity$_{t-1}$/Total liabilities$_{t-1}$)

Results

		Predicted Sign	Dependent Variable: *ue_cfo* Panel A	Panel B
H1	ivo_t	+	0.295***	0.331***
			(19.214)	(21.265)
	ivi_t	-	-0.408***	-0.463***
			(-5.868)	(-6.696)
	$fino_t$	+	0.410***	0.407***
			(11.556)	(11.444)
	$fini_t$	-	-0.250***	-0.273***
			(-21.501)	(-23.100)
	$zscore_{t-1}$	?	-0.000	-0.000
			(-0.696)	(-0.025)
H3	$zscore_{t-1}*ivo_t$	+	0.004**	0.005***
			(2.052)	(2.715)
	$zscore_{t-1}*ivi_t$	-	-0.005	-0.007
			(-0.659)	(-0.926)
	$zscore_{t-1}*fino_t$	+	0.009	0.008
			(1.464)	(1.309)
	$zscore_{t-1}*fini_t$	-	-0.002*	-0.003**
			(-1.790)	(-2.412)
	roa_t	+	0.514***	0.550***
			(40.789)	(42.913)
	$size_t$	-	-0.003***	-0.003***
			(-5.669)	(-4.280)
	mtb_t	+	0.001	0.001*
			(0.986)	(1.931)
	$dacc_t$	-	-0.653***	-0.656***
			(-76.284)	(-75.130)
	Constant		-0.034***	-0.045***
			(-7.252)	(-5.277)
	Industry dummies		No	Yes
	Year dummies		No	Yes
	No. of observations		13305	13305
	Adjusted R-Square		53.3%	54.7%
	p-value		0.000	0.000

		Predicted Sign	Dependent Variable: *ue_cfo*	
			Panel A	Panel B
H1	cff_t	-	-0.275***	-0.293***
			(-25.891)	(-26.716)
	cfi_t	-	-0.305***	-0.340***
			(-21.124)	(-23.047)
	$zscore_{t-1}$	?	0.000	0.000
			(0.570)	(1.008)
H3	$zscore_{t-1}*cff_t$	-	-0.004***	-0.004***
			(-2.941)	(-3.372)
	$zscore_{t-1}*cfi_t$	-	-0.004**	-0.005***
			(-2.362)	(-3.031)
	roa_t	+	0.527***	0.563***
			(42.757)	(45.003)
	$size_t$	-	-0.003***	-0.003***
			(-6.144)	(-4.495)
	mtb_t	+	0.001	0.001**
			(1.041)	(2.163)
	$dacc_t$	-	-0.662***	-0.665***
			(-78.092)	(-77.336)
	Constant		-0.026***	-0.039***
			(-6.048)	(-4.679)
	Industry dummies		No	Yes
	Year dummies		No	Yes
	No. of observations		13305	13305
	Adjusted R-Square		53.1%	54.6%
	p-value		0.000	0.000

Rs. 4.40 Crores
Rs. 21.60 Crores

Additional
Rs. 6 Lakhs
Rs. 31.70 Lakhs

0.73% (3.58%)
of Sales

0.57% (2.79%)
of Assets

	Predicted Sign	Dependent Variable: *ue_cfo*	
		Panel A	Panel B
ivo_t	+	0.302***	0.315***
		(80.083)	(83.652)
ivi_t	-	-0.424***	-0.513***
		(-44.099)	(-53.917)
fin_t	+	0.614***	0.651***
		(59.373)	(64.040)
$fini_t$	-	-0.228***	-0.240***
		(-89.824)	(-95.747)
$zscore_{t-1}$	?	-0.000	0.000***
		(-1.117)	(4.305)
$zscore_{t-1}*ivo_t$	+	-0.002***	-0.001***
		(-14.201)	(-12.111)
$zscore_{t-1}*ivi_t$	-	-0.005***	-0.006***
		(-10.996)	(-11.870)
$zscore_{t-1}*fino_t$	+	0.007***	0.007***
		(7.444)	(7.941)
$zscore_{t-1}*fini_t$	-	0.001***	0.001***
		(14.464)	(12.195)
ind_t	?	-0.032***	-0.022
		(-16.278)	(-0.000)
H2 ind_t*ivo_t	+	0.302***	0.363***
		(16.338)	(19.574)
ind_t*ivi_t	-	-0.532***	-0.575***
		(-5.929)	(-6.530)
ind_t*fino_t	+	0.241***	0.291***
		(5.648)	(6.873)
ind_t*fini_t	-	-0.277***	-0.316***
		(-18.803)	(-21.677)
H4 $ind_t*zscore_{t-1}*ivo_t$	+	0.012***	0.016***
		(5.192)	(6.933)
$ind_t*zscore_{t-1}*ivi_t$	-	-0.010	-0.011
		(-0.991)	(-1.139)
$ind_t*zscore_{t-1}*fino_t$	+	0.003	0.011
		(0.427)	(1.487)
$ind_t*zscore_{t-1}*fini_t$	-	-0.010***	-0.012***
		(-5.533)	(-6.920)
roa_t	+	0.191***	0.213***
		(84.812)	(94.556)
$size_t$	-	-0.006***	-0.006***
		(-27.120)	(-26.743)
mtb_t	+	0.002***	0.001***
		(16.028)	(9.941)
$dacc_t$	-	-0.008***	-0.010***
		(-13.715)	(-17.248)
Constant		0.012***	-0.024
		(9.007)	(-1.247)
Industry dummies		No	Yes
Year dummies		No	Yes
No. of observations		89563	89563
Adjusted R-Square		37.6%	40.7%
p-value		0.000	0.000

		Predicted Sign	Dependent Variable: *ue_cfo* Panel A	Panel B
	cff_t	-	-0.252***	-0.268***
			(-104.346)	(-112.067)
	cfi_t	-	-0.314***	-0.339***
			(-92.131)	(-99.514)
	$zscore_{t-1}$	?	-0.000	0.000***
			(-1.617)	(4.624)
	$zscore_{t-1}*cff_t$	-	0.001***	0.000***
			(11.522)	(9.184)
	$zscore_{t-1}*cfi_t$	-	0.001***	0.001***
			(10.671)	(8.089)
	ind_t	?	-0.033***	0.041
			(-20.248)	(0.000)
	ind_t*cff_t	-	-0.332***	-0.369***
H2			(-25.401)	(-28.174)
	ind_t*cfi_t	-	-0.342***	-0.395***
			(-19.884)	(-22.585)
	$ind_t*zscore_{t-1}*cff_t$	-	-0.013***	-0.015***
H4			(-7.879)	(-9.478)
	$ind_t*zscore_{t-1}*cfi_t$	-	-0.013***	-0.017***
			(-6.291)	(-8.044)
	roa_t	+	0.191***	0.212***
			(84.165)	(93.449)
	$size_t$	-	-0.006***	-0.006***
			(-26.392)	(-25.820)
	mtb_t	+	0.002***	0.001***
			(17.616)	(11.826)
	$dacc_t$	-	-0.008***	-0.010***
			(-13.719)	(-17.158)
	Constant		0.022***	-0.014
			(16.883)	(-0.718)
	Industry dummies		No	Yes
	Year dummies		No	Yes
	No. of observations		89563	89563
	Adjusted R-Square		36.4%	39.4%
	p-value		0.000	0.000

Conclusion

- Managers of Indian firms consider operating cash flows as an important indicator and manipulate this figure
- Such manipulation is higher in India vis-à-vis that in the United States
- Financially distressed firms in India are more likely to misclassify cash flows
- Cash flows are as prone to manipulation and misclassification as the earnings
- Weaker corporate governance, investor protection and less regulatory oversight seem to be the reasons